Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Full-stack software engineer with 2+ years building and maintaining **React**, **TypeScript**, Python, and Node.js applications. Experienced across concept, design, implementation, testing, deployment, production support, visual AI workflows, and user-centered development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and maintained a multi-tenant **React** and Python platform for 500+ users, optimizing user workflows, indexed queries, and backend access paths to reduce application latency from 1.2 seconds to 350 milliseconds.
- Owned application changes through implementation, testing, deployment, and post-launch support for 5,000+ monthly jobs, adding validation, monitoring, error recovery, and CI/CD releases that sustained greater than 99% reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python Flask services handling 10,000+ daily requests, translating product requirements into upload, history, status, and investigation workflows while maintaining responsive behavior during peak load.
- Collaborated with users and engineers to refine functional designs, then implemented validation, progress feedback, defect recovery, and testable processing for 1,000+ files, reducing end-to-end turnaround by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **visual-data pipeline** processing 100,000+ sensor and trajectory records, improving model quality 18% through modular implementation, reproducible experiments, automated verification, performance tracing, and structured configuration.
- Implemented spatial-temporal modeling that combined map geometry with vehicle sensor trajectories, improving closed-loop stability 15% while supporting systematic debugging, feasibility analysis, and comparative evaluation across complex visual scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered enterprise integration software with automated builds and VM-based validation across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito.
- Developed object-oriented microservices and AWS data pipelines using Glue, S3, SQL, caching, and pagination, improving response times 30%, processing efficiency 68%, and maintainability across integrated platforms with reliable operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, TypeScript, and Node.js features across eight microservices, integrating WebSockets and REST APIs to synchronize more than two million records across responsive customer and operational workflows.
- Automated testing and cloud delivery with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving release validation and production observability across environments.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
Frontend: React.js, Next.js, TypeScript, JavaScript, HTML, CSS, Material UI, Tailwind CSS, responsive design, component architecture, user-centered interfaces
Backend & Data: Python, Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, microservices, PostgreSQL, MySQL, MongoDB, Azure SQL
Visual AI: PyTorch, Transformers, image segmentation, computer vision workflows, multimodal AI, sensor data, metadata processing, model evaluation
Delivery & Tools: Git, GitHub, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, AWS, Azure, GCP, Sentry, logging, monitoring
Engineering Practices: Agile, SDLC, UX collaboration, technical feasibility, unit testing, integration testing, debugging, code reviews, maintenance, production support, documentation

Projects

PiXMod | Live | *Next.js, PyTorch, Transformers, MongoDB*
- Built a visual AI application using Next.js, Transformers, and MongoDB for segmentation, retrieval, and generative editing, processing 1,000+ image assets through responsive workflows, metadata storage, and cloud deployment.

HMIS | Live | *React, TypeScript, REST APIs, Sentry*
- Built a role-based web application with scheduling, dashboards, notifications, validation, and Sentry monitoring, reducing administrative workload 30% while supporting secure access, maintainable components, and reliable post-launch operations.

Big-Brain | Live | *Python, LLaMA-2, FAISS, PostgreSQL, GCP*
- Built an AI web platform processing 50,000+ records with responsive search, summarization, secure access, and incremental indexing, reducing retrieval latency 40% through optimized backend services and data pipelines.